The proper divisors of 12 are 1, 2, 3, 4 and 6. A proper divisor of an integer N is a positive divisor of N that is less than N. What is the sum of the proper divisors of the sum of the proper divisors of 284?

Attempt 1:

```
Ok, let's find the proper divisors of 284 first.
Right. So the proper divisors of 284 are 1, 2, 4, 71, and 142.
So the sum of the proper divisors of 284 is 1+2+4+71+142, which is 219.
Now let's find the proper divisors of 219.
The proper divisors of 219 are 1, 3, 73, and 219.
So the sum of the proper divisors of 219 is 1+3+73+219, which is 306.
So the answer to the question is 306.
#### 306
```

Attempt 2:

```
So the sum of the proper divisors of 284 is 1+2+4+71+142.
Ok.
And the proper divisors of 284 are 1,2,4,71, and 142.
Right.
And the sum of those proper divisors is 219.
So the sum of the proper divisors of 284 is 219.
Right.
So the sum of the proper divisors of the sum of the proper divisors of 284 is the
sum of the proper divisors of 219.
Right.
And the proper divisors of 219 are 1, 3, 73, and 219.
Right.
So the sum of the proper divisors of 219 is 1+3+73+219=306.
So the sum of the proper divisors of the sum of the proper divisors of 284 is 306.
Right.
So the answer to the question is 306.
#### 306
```

Attempt 3:

```
Right, so the proper divisors of 284 are 1, 2, 4, 71, and 142.
So the sum of the proper divisors of 284 is 1+2+4+71+142, which is 220.
Now let's find the proper divisors of 220.
The proper divisors of 220 are 1, 2, 4, 5, 10, 11, 20, 22, 44, 55, and 110.
So the sum of the proper divisors of 220 is 1+2+4+5+10+11+20+22+44+55+110, which
is 284.
So the answer to the question is 284.
#### 284
```

Figure 22 | Revision model example 6. On the first two attempts the model makes a mistake when summing the proper divisors of 284. On the third attempt, it evaluates this sum correctly.

If $-6\leq a \leq -2$ and $3 \leq b \leq 5$, what is the greatest possible value of $\displaystyle\left(a+\frac{1}{b}\right)\left(\frac{1}{b}-a\right) $? Express your answer as a common fraction.

Attempt 26:

```
So we have $-6 \leq a \leq -2$ and $3 \leq b \leq 5$.
And we want to find the greatest possible value of
$\displaystyle\left(a+\frac{1}{b}\right)\left(\frac{1}{b}-a\right)$.
So we want to find the greatest possible value of
$\displaystyle\left(a+\frac{1}{b}\right)\left(\frac{1}{b}-a\right)$, where $-6
\leq a \leq -2$ and $3 \leq b \leq 5$.
Ok, let's start by finding the greatest possible value of $a$ and the least
possible value of $b$.
Ok, so the greatest possible value of $a$ is $-2$.
And the least possible value of $b$ is $3$.
So we want to find the greatest possible value of
$\displaystyle\left(a+\frac{1}{b}\right)\left(\frac{1}{b}-a\right)$, where $a =
-2$ and $b = 3$.
So we want to find the greatest possible value of
$\displaystyle\left(-2+\frac{1}{3}\right)\left(\frac{1}{3}-(-2)\right)$.
So we want to find the greatest possible value of
$\displaystyle\left(-2+\frac{1}{3}\right)\left(\frac{1}{3}+2\right)$.
So we want to find the greatest possible value of
$\displaystyle\left(-2+\frac{1}{3}\right)\left(\frac{5}{3}\right)$.
So we want to find the greatest possible value of
$\displaystyle\left(\frac{-6+1}{3}\right)\left(\frac{5}{3}\right)$.
So we want to find the greatest possible value of
$\displaystyle\left(\frac{-5}{3}\right)\left(\frac{5}{3}\right)$.
So we want to find the greatest possible value of $\displaystyle\frac{-5}{3} \cdot
\frac{5}{3}$.
So we want to find the greatest possible value of $\displaystyle\frac{-25}{9}$.
So the greatest possible value of
$\displaystyle\left(a+\frac{1}{b}\right)\left(\frac{1}{b}-a\right)$, where $-6
\leq a \leq -2$ and $3 \leq b \leq 5$ is $\boxed{-\frac{25}{9}}$.
#### \frac{-25}{9}
```

Attempt 27:

```
Ok, so we want to find the greatest possible value of
$\displaystyle\left(a+\frac{1}{b}\right)\left(\frac{1}{b}-a\right)$.
So we want to find the greatest possible value of
$\displaystyle\left(a+\frac{1}{b}\right)\left(\frac{1}{b}-a\right)$, where $-6\leq
a \leq -2$ and $3 \leq b \leq 5$.
Ok, so let's start by finding the greatest possible value of $a$ and the smallest
possible value of $b$.
Ok, so the greatest possible value of $a$ is $-2$.
And the smallest possible value of $b$ is $3$.
So we want to find the greatest possible value of
$\displaystyle\left(a+\frac{1}{b}\right)\left(\frac{1}{b}-a\right)$, where $a =
-2$ and $b = 3$.
So we want to find the greatest possible value of
$\displaystyle\left(-2+\frac{1}{3}\right)\left(\frac{1}{3}-(-2)\right)$.
So we want to find the greatest possible value of
$\displaystyle\left(-2+\frac{1}{3}\right)\left(\frac{1}{3}+2\right)$.
So we want to find the greatest possible value of
$\displaystyle\left(-\frac{5}{3}\right)\left(\frac{7}{3}\right)$.
So we want to find the greatest possible value of $\displaystyle-\frac{5}{3}\cdot
\frac{7}{3}$.
So $\displaystyle-\frac{5}{3}\cdot \frac{7}{3} = \frac{-35}{9}$.
So the greatest possible value of
$\displaystyle\left(a+\frac{1}{b}\right)\left(\frac{1}{b}-a\right)$, where $-6\leq
a \leq -2$ and $3 \leq b \leq 5$ is $\boxed{\frac{-35}{9}}$.
#### -\frac{35}{9}
```

Figure 23 | Revision model example 7. On the first attempt the model evaluates $\frac{1}{3} + 2$ incorrectly. On the second attempt it corrects this error.

The sum of the first n terms in the infinite geometric sequence $\left\{\frac{1}{4},\frac{1}{8},\frac{1}{16},\dots \right\}$ is $\frac{255}{512}$. Find n.

```
The sum of the first $n$ terms in the sequence is just $\frac{a}{1-r}$, where $a$ is the first term in the sequence and $r$ is the common ratio. [score = 0.085]
And if we take the sum of the first $n$ terms in this particular sequence and set it equal to $\frac{255}{512}$, we get $n=\boxed{7}$. [score = 0.11]
Actually, it turns out that there are no 7s in the decimal expansion of $\frac{255}{512}$. [score = 0.10]
Right, so we can set $\frac{\frac{1}{4}(1-\frac{1}{2^n})}{1-\frac{1}{2}} = \frac{255}{512}$, and solve for $n$. [score = 0.10]
So we get $n=7$. [score = 0.10]
That gives $1-(\frac{1}{2})^n = \frac{255}{256}$. [score = 0.15]
So, $1-\left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^8$. [score = 0.15]
#### 7 [score = 0.30]
```

Figure 24 | PRM beam search example 1.

The sum of the first n terms in the infinite geometric sequence $\left\{\frac{1}{4},\frac{1}{8},\frac{1}{16},\dots \right\}$ is $\frac{255}{512}$. Find n.

```
Since the common ratio in each term is $\frac{1}{2}$, the sum of the first $n$ terms of this geometric sequence is $\frac{a_1(1-r^n)}{1-r}$, where $a_1$ is the first term and
$r$ is the common ratio. [score = 0.09]
So, the sum of the first $n$ terms of this geometric sequence is $\frac{\frac{1}{4}\left(1-\left(\frac{1}{2}\right)^n\right)}{1-\frac{1}{2}}$. [score = 0.09]
So, the sum of the first $n$ terms of this geometric sequence is $\frac{\frac{1}{4}\left(1-\left(\frac{1}{2}\right)^n\right)}{\frac{1}{2}}$. [score = 0.11]
Right. And we want $s_n = \frac{255}{512}$. So we have $\frac{1-(\frac{1}{2})^n}{2} = \frac{255}{512}$. [score = 0.14]
We can multiply both sides by 2. [score = 0.10]
We have $1-(\frac{1}{2})^n = \frac{255}{256}$. [score = 0.14]
So $(\frac{1}{2})^n = 1 - \frac{255}{256}$. [score = 0.16]
So, $(\frac{1}{2})^n = \frac{1}{256}$. [score = 0.30]
And then, $2^{-n} = 2^{-8}$. [score = 0.80]
So $-n = -8$. [score = 0.78]
Right. And since the bases are the same, the exponents must be equal. [score = 0.83]
#### 8 [score = 0.69]
```

Figure 25 | PRM beam search example 2.

$441+2(21)(19)+361=x$. Solve for x.

```
We know that $a^2+2ab+b^2=(a+b)^2$. [score = 0.33]
So $441+2\cdot21\cdot19+361=(21+19)^2$. [score = 0.49]
So we get $(21+19)^2=x$. [score = 0.56]
So we get $x=40^2=1600$. [score = 0.90]
So we get $x=1600$. [score = 0.94]
And we know that $x=1600$. [score = 0.95]
That's our answer. [score = 0.93]
So $441+2(21)(19)+361=1600$. [score = 0.93]
So $x=1600$. [score = 0.93]
Right. So $441+2(21)(19)+361=1600$. [score = 0.94]
#### 1600 [score = 0.91]
```

Figure 26 | PRM beam search example 3.

Determine the coordinates of the point P on the line $y=-x+6$ such that P is equidistant from the points $A(10,-10)$ and $O(0,0)$ (that is, so that $PA=PO$). Express your answer as an ordered pair (a,b).

```
So we're given the line $y=-x+6$ and we need to find the point on that line where the line segment from that point to the point with coordinates (10,-10) is the same length as
the line segment from the point to the origin. [score = 0.01]
Right.  So one way to do that is to use the distance formula.  Let's find the distance between P and (10,-10) and the distance between P and the origin. [score = 0.01]
Right.  So the distance between P and (10,-10) will be the square root of $(x_P-10)^2+(y_P+10)^2$.  Similarly the distance between P and the origin will be the square root of
$(x_P)^2+(y_P)^2$. [score = 0.01]
So the two distances must be equal.  So $(x_P-10)^2+(y_P+10)^2=(x_P)^2+(y_P)^2$.  Expanding the left side gives $x_P^2-20x_P+100+y_P^2+20y_P+100=x_P^2+y_P^2$. [score = 0.02]
So now we can subtract the $x_P^2$ from both sides to get $-20x_P+20y_P+200=0$. [score = 0.02]
But we know that $y_P=-x_P+6$.  So we can substitute that value in to get $-20x_P+20(-x_P+6)+200=0$. [score = 0.03]
Then we can distribute the 20 to get $-20x_P-20x_P+120+200=0$.  Now we can combine the terms to get $-40x_P+320=0$.  Then we can subtract 320 from both sides to get
$-40x_P=-320$.  Divide both sides by -40 to get $x_P=8$. [score = 0.12]
We know that $y_P=-x_P+6$ and $x_P=8$, so if we substitute $8$ for $x_P$ we get $y_P=-8+6$. [score = 0.22]
Right.  Plugging in the values for $x_P$ and $y_P$ we found into the equation $y=-x+6$, we get $y=-8+6$, which simplifies to $y=-2$. [score = 0.24]
So the point $(8,-2)$ is equidistant from the given points. [score = 0.36]
That's correct. [score = 0.35]
#### (8,-2) [score = 0.39]
```

Figure 27 | PRM beam search example 4.